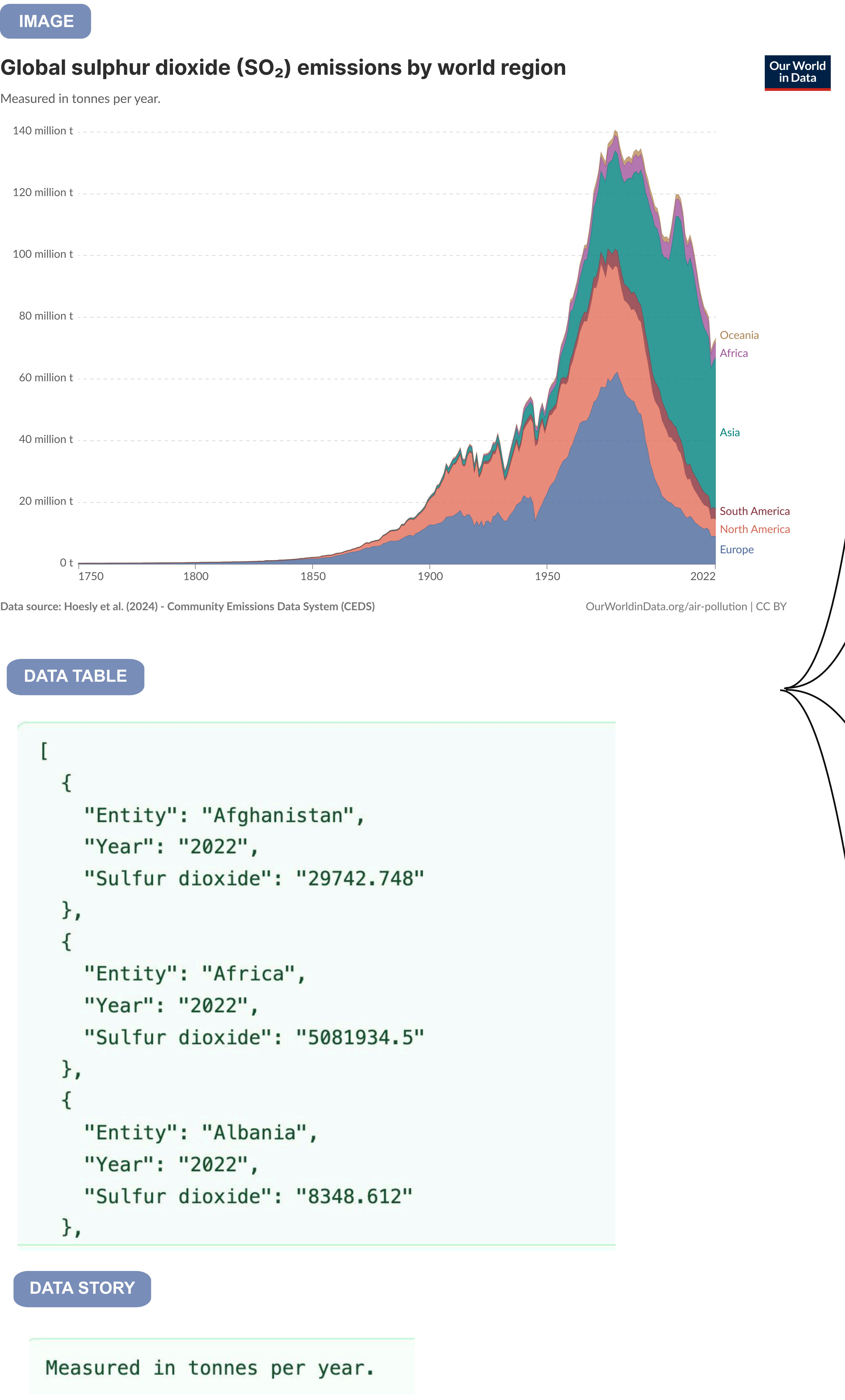
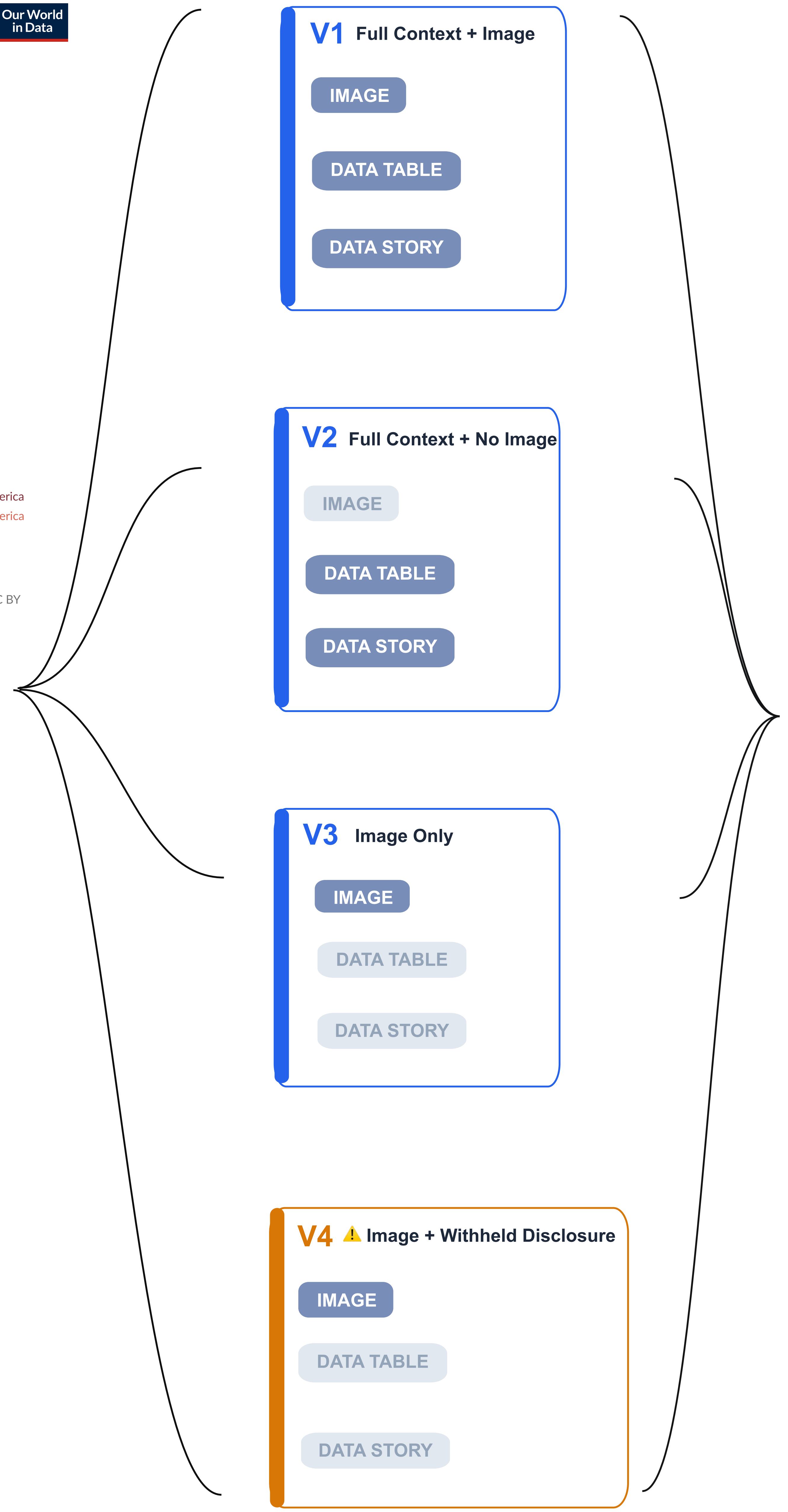
(A) Source Inputs Collected



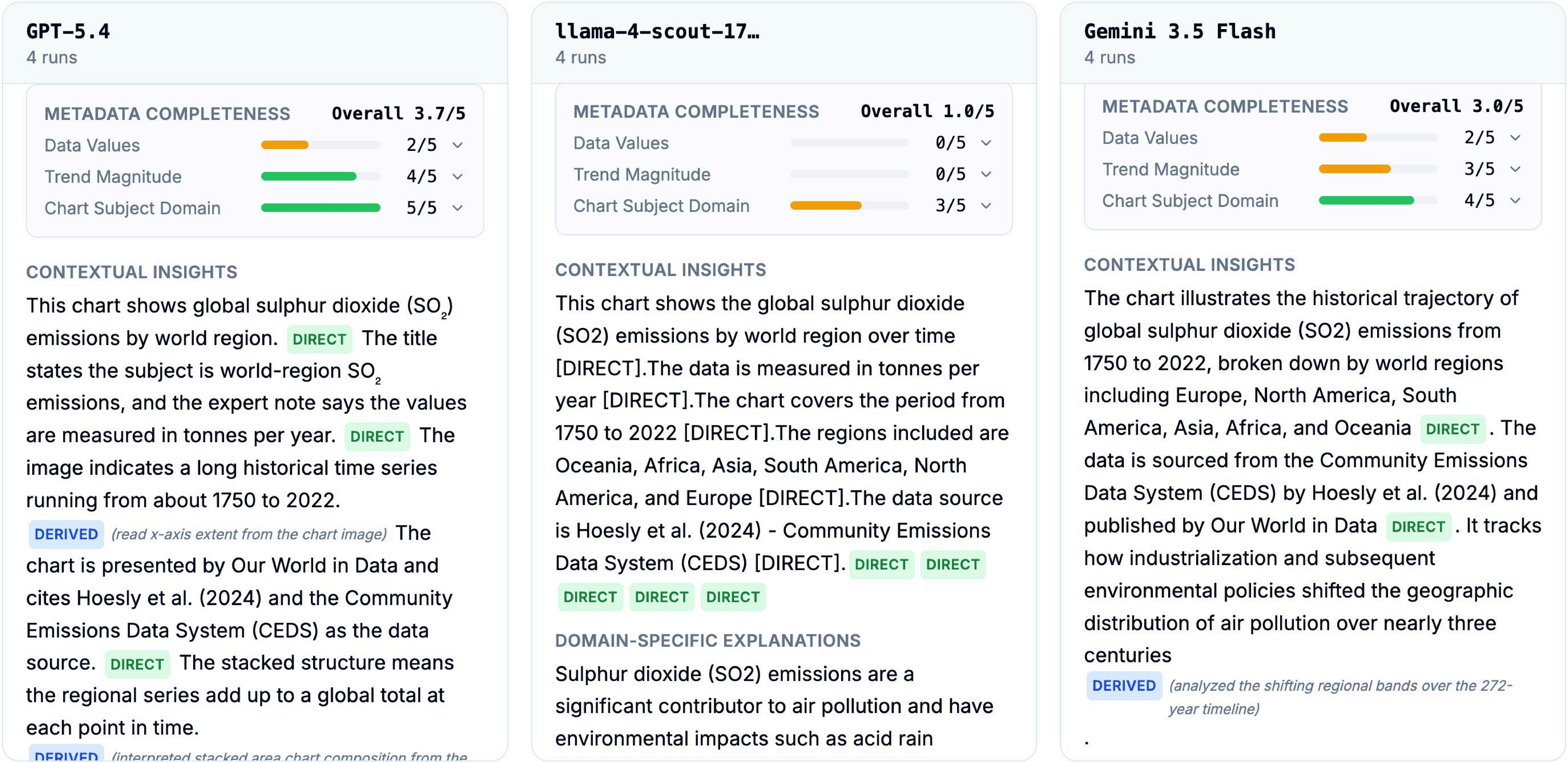
```
[
  {
    "Entity": "Afghanistan",
    "Year": "2022",
    "Sulfur dioxide": "29742.748"
  },
  {
    "Entity": "Africa",
    "Year": "2022",
    "Sulfur dioxide": "5081934.5"
  },
  {
    "Entity": "Albania",
    "Year": "2022",
    "Sulfur dioxide": "8348.612"
  },
```

```
Measured in tonnes per year.
```

(B) MLLM Input Conditions



(C) MLLM Generated Texts with Model-attributed Source labels



GPT-5.4
4 runs

METADATA COMPLETENESS **Overall 3.7/5**

Data Values	2/5	⌄
Trend Magnitude	4/5	⌄
Chart Subject Domain	5/5	⌄

CONTEXTUAL INSIGHTS

This chart shows global sulphur dioxide (SO_2) emissions by world region. DIRECT The title states the subject is world-region SO_2 emissions, and the expert note says the values are measured in tonnes per year. DIRECT The image indicates a long historical time series running from about 1750 to 2022.

DERIVED *(read x-axis extent from the chart image)* The chart is presented by Our World in Data and cites Hoesly et al. (2024) and the Community Emissions Data System (CEDS) as the data source. DIRECT The stacked structure means the regional series add up to a global total at each point in time.

DERIVED *(interpreted stacked area chart composition from the*

llama-4-scout-17…
4 runs

METADATA COMPLETENESS **Overall 1.0/5**

Data Values	0/5	⌄
Trend Magnitude	0/5	⌄
Chart Subject Domain	3/5	⌄

CONTEXTUAL INSIGHTS

This chart shows the global sulphur dioxide (SO2) emissions by world region over time [DIRECT].The data is measured in tonnes per year [DIRECT].The chart covers the period from 1750 to 2022 [DIRECT].The regions included are Oceania, Africa, Asia, South America, North America, and Europe [DIRECT].The data source is Hoesly et al. (2024) - Community Emissions Data System (CEDS) [DIRECT]. DIRECT DIRECT DIRECT DIRECT DIRECT

DOMAIN-SPECIFIC EXPLANATIONS

Sulphur dioxide (SO2) emissions are a significant contributor to air pollution and have environmental impacts such as acid rain

Gemini 3.5 Flash
4 runs

METADATA COMPLETENESS **Overall 3.0/5**

Data Values	2/5	⌄
Trend Magnitude	3/5	⌄
Chart Subject Domain	4/5	⌄

CONTEXTUAL INSIGHTS

The chart illustrates the historical trajectory of global sulphur dioxide (SO2) emissions from 1750 to 2022, broken down by world regions including Europe, North America, South America, Asia, Africa, and Oceania DIRECT . The data is sourced from the Community Emissions Data System (CEDS) by Hoesly et al. (2024) and published by Our World in Data DIRECT . It tracks how industrialization and subsequent environmental policies shifted the geographic distribution of air pollution over nearly three centuries

DERIVED *(analyzed the shifting regional bands over the 272-year timeline)*

.